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Thinking and Programming
Semester- odd
Batch- ALL

Tutorial Assignment: 5

Tutorial title: Iterative Control and Loops

CO Mapping

Name	CO1	CO2	CO3
Iterative Control and Loops	✓	-	-

CONCEPT CLARITY:

Python has two primitive loop commands:

- while loops
- for loops

While loop Syntax

```
while expression:  
    statement(s)
```

While loop with else

```
while condition:  
    # execute these statements  
else:  
    # execute these statements
```

Nested Loops

```
while expression:  
    while expression:  
        statement(s)  
        statement(s)
```

For loop in python acts as “for in” loop

For loop Syntax

```
for iterator_var in sequence:  
    statements(s)
```

For loop with else

```
for iterator_var in sequence:  
    # execute these statements  
else:  
    # execute these statements
```

```
for iterator_var in sequence:  
    for iterator_var in sequence:  
        statements(s)  
        statements(s)
```

Conditional Statements

Continue Statement

It returns the control to the beginning of the loop.

Break Statement

It brings control out of the loop

Sequence Operators

Concatenation : +

Repetition : *

Membership Testing : x in S

Slicing a Sequence : $S[i:j]$, $S[i:]$, $S[:j]$

List Methods

Method	Description
<code>L.count(x)</code>	Returns the number of items of L that are equal to x .
<code>L.index(x)</code>	Returns the index of the first occurrence of an item in L that is equal to x , or raises an exception if L has no such item.
<code>L.append(x)</code>	Appends item x to the end of L ; e.g., $L[\text{len}(L):]=[x]$.
<code>L.extend(s)</code>	Appends all the items of iterable s to the end of L ; e.g., $L[\text{len}(L):]=s$.
<code>L.insert(i, x)</code>	Inserts item x in L before the item at index i , moving following items of L (if any) "rightward" to make space (increases $\text{len}(L)$ by one, does not replace any item, does not raise exceptions: acts just like $L[i:i]=[x]$).
<code>L.remove(x)</code>	Removes from L the first occurrence of an item in L that is equal to x , or raises an exception if L has no such item.
<code>L.pop([i])</code>	Returns the value of the item at index i and removes it from L ; if i is omitted, removes and returns the last item; raises an exception if L is empty or i is an invalid index in L .
<code>L.reverse()</code>	Reverses, in place, the items of L .
<code>L.sort([f])</code> (2.3)	Sorts, in place, the items of L , comparing items pairwise via function f ; if f is omitted, comparison is via the built-in function <code>cmp</code> . For more details, see Sorting a list .
<code>L.sort(cmp=cmp, key=None, reverse=False)</code> (2.4)	Sorts, in-place, the items of L , comparing items pairwise via the function passed as <code>cmp</code> (by default, the built-in function <code>cmp</code>). When argument <code>key</code> is not <code>None</code> , what gets compared for each item x is <code>key(x)</code> , not x itself. For more details, see Sorting a list .

EXERCISE

Q1: Draw the flowchart for “While loop” in Conditional Statements

Q2: Predict the output

```
i = 1
while (i ==1):
    print (i)
```

Q3: If the code below is correct, write its output

```
string = 'BENNETT'
for letters in string:
    print (letters,end='X')
```

Q4: predict the error

```
num1 = 17
num2 = -12
while num1 > 5 and num2 < -5 :
    num1 -= 2
    num2 += 3
    print( (num1, num2) )
```

Q5: predict the output

```
for string in "BU Student":
    pass
print(string)
```

Q6: predict the output

```
a = [1,2,3,4,5]x
= []
while a:
    x.append( (a.pop())**2)
print( x )
```

- a. [1,2,3,4,5]
- b. [5,4,3,2,1]
- c. [25,16,9,4,1]
- d. [1,4,9,16,25]

Q7: predict the last digit in the output of the program below

```
X = []
i = 0
```

```
while len(X) < 10 :  
    X.append(i)  
    i += 2  
print(X)
```

- a. 10
- b. 20
- c. 9
- d. 18

Q8: Write a program to add two numbers multiple times in shell only, without using run again and again.

Q9: Make a BMI calculator that shows all four conditions: underweight, overweight, normal and obese.

Q10: Make a program to obtain sum and average of a sequence of numbers that starts from a given number and ends at a predefined proceeding step.:

Eg: 1+2+3+4: this sequence starts from 1 and taking three proceeding steps

Q11: WAP to find the shortest coordinates from the origin in 2D system

Q12: Write a program to see a table of a given number.

Q13: WAP to count number of letters in a string. Do not count spaces.

Q14: WAP to number of count digits in a string. Do not count spaces.

Q15: WAP to number of count digits and letters in a string. Do not count spaces.

Bonus question:

WAP for a bakery shop management system. Consider the shop contains 10 items. You must print the bill that shows cost per item as well as total cost. Note that items should be defined by a key not by its name to reduce the time taken to generate a bill.